| Question | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| 1 | *(a)* | (i) | Capacitor equation rearranged $d = \frac{\varepsilon_0 A}{C}$ (1)<br>Substitution of values (1)<br>Correct answer $1.43 \times 10^{-4}$ m (0.143 mm) (1)<br>[Accept $1.4\times10^{-4}$ m, not $1\times10^{-4}$] | 1 | 1<br>1 | | 3 | 3 | |
| | | (ii) | Adding capacitors in parallel (4 nF) (1)<br>Understanding of 6 nF in series with 4 nF i.e. correct application of physics (1) [even failure to invert to get answer, so 1/4 + 1/6 $\rightarrow$ 2 marks]<br>Correct answer 2.4 n[F] or 12/5 n[F] (1) c.a.o. | 1 | 1<br>1 | | 3 | 2 | |
| | | (iii) | Any valid *CV* e.g. 2 × 9, 6 × 6, 2.4 × 15 (1)<br>[Accept 2.4 × 15 or ½ × 2.4 × 15; **not** 1.2 × 15 or 2.4 × 7.5]<br>½-way explanation: charge split / divided between 2 nFs / 9 V across 2 nF / 6 V across 6 nF (1)<br>Completed explanation: all charge on capacitors in series / justify 9 V or 6V [e.g. V $\propto$ 1/C] | 1 | 1<br>1 | | 3 | 2 | |
| | | (iv) | <u>Correct</u> substitution (into valid equation) e.g.0.5 $QV$ etc. (1)<br>Correct answer (270 nJ or $2.7 \times 10^{-7}$J) (1) | 1 | 1 | | 2 | 2 | |
| | *(b)* | (i) | Capacitor in series with resistor and cell/battery/psu (1)<br>Properly placed ammeter and voltmeter (1) | 2 | | | 2 | | 2 |

| Question | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| | | (ii) | Emf = 12 V stated [accept $V_{max}$] (1)<br>Initial current = 2.7 mA (1)<br>$R = \frac{V}{I}$ = 4 444 Ω (1) [4.4 kΩ]<br>[N.B. With incompatible *V* and *I*, 2nd and 3rd marks not accessible]<br>$RC$ ~ 8 s from 37% $I_{max}$ value or 63% $V_{max}$ value or intercept of gradient at the origin with $I$ = 0 (or $V$ = 12 V) **OR** taking logs and rearranging **OR** obtaining $Q$ from total area under current graph around 2.0 × $10^{-2}$ C (1)<br>$C = \frac{8.0}{4400}$ = 1.8 [± 0.3] mF (1) [No ecf]<br>**Unit** penalty (–1): all 3 units [V, Ω, F] needed<br><br>**Alternative**<br>Emf = 12 V (1)<br>Find $\frac{dV}{dt}$ **and** $I$ at given $t$, e.g. at 10 s, 0.411 V $s^{-1}$, 7.5 mA (1)<br>Hence calculate *C* from $I = C\frac{dV}{dt}$ → 1.8 mF ecf (1)<br>$RC$ ~ 8 s from 37% $I_{max}$ value or 63% $V_{max}$ value or intercept of gradient at *t* = 0 with $I$ = 0 (or $V$ = 12 V) (1)<br>*R* from *C* and *RC* → 4.4 kΩ (1)<br>**Unit** penalty (–1): all 3 units [V, Ω, F] needed | | 5 | | 5 | 5 | 5 |
| | | | **Question 1 total** | **6** | **12** | **0** | **18** | **14** | **7** |